Question 46: End-of-Day Ledger Posting Simulation

Problem Statement

A bank batch simulates posting a set of credits and debits to a ledger.

Task

Read an opening balance and a sequence of transaction IDs, types, and amounts. Apply valid credits and debits in order. Reject negative amounts, unknown types, and debits that would overdraw the balance. Print each transaction result and the final balance.

Input Format

- Line 1: Opening balance.
- Line 2: Number of transactions (N).
- Next N lines: Transaction ID, type (C or D), and amount.

Output Format

Print each transaction's POSTED or REJECTED status and the final balance.

Constraints

- 0 <= opening balance <= 100,000,000
- 1 <= N <= 100,000
- Amounts have two decimal places.

2. Approach

- Read the opening balance and the number of transactions
- Process each transaction in the given order using its transaction ID, type, and amount.
- If the amount is negative or the transaction type is invalid, reject the transaction.
- If the type is C, add the amount to the balance and mark it as POSTED.
- If the type is D, check whether the amount exceeds the current balance. If it does, reject the transaction; otherwise, subtract the amount and mark it as POSTED.
- Print each transaction's status and the final balance to two decimal places.

3. Algorithm

1. Start.
2. Read the opening balance.
3. Read the number of transactions N
4. Repeat for each transaction from 1 to N:
    - Read transaction ID, type, and amount.
    - If the amount is negative or the type is neither C nor D, print REJECTED.
    - Else, if the type is C, add the amount to the balance and print POSTED.

- Else, if the type is D and the amount is greater than the balance, print REJECTED.
- Otherwise, subtract the amount from the balance and print POSTED.

5. Print the final balance to two decimal places.
6. Stop.

4. COBOL Program

```
IDENTIFICATION DIVISION.
PROGRAM-ID. LEDGER.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 BALANCE-AMT PIC 9(9)V99.
01 N PIC 9(6).
01 I PIC 9(6).
01 TRANS-ID PIC X(20).
01 TRANS-TYPE PIC X.
01 AMOUNT PIC S9(9)V99.
01 DISPLAY-BALANCE PIC Z(8)9.99.

PROCEDURE DIVISION.
   ACCEPT BALANCE-AMT
   ACCEPT N

   PERFORM VARYING I FROM 1 BY 1
      UNTIL I > N
      ACCEPT TRANS-ID
      ACCEPT TRANS-TYPE
      ACCEPT AMOUNT

      IF AMOUNT < 0
         DISPLAY TRANS-ID " REJECTED"
      ELSE
         IF TRANS-TYPE = 'C'
```

```
            ADD AMOUNT TO BALANCE-AMT
            DISPLAY TRANS-ID " POSTED"
          ELSE
            IF TRANS-TYPE = 'D'
              IF AMOUNT > BALANCE-AMT
                DISPLAY TRANS-ID " REJECTED"
              ELSE
                SUBTRACT AMOUNT
                  FROM BALANCE-AMT
                DISPLAY TRANS-ID " POSTED"
              END-IF
            ELSE
              DISPLAY TRANS-ID " REJECTED"
            END-IF
          END-IF
        END-IF
      END-PERFORM
      MOVE BALANCE-AMT TO DISPLAY-BALANCE
      DISPLAY "Final balance: " DISPLAY-BALANCE
      STOP RUN.
```

5. Sample Input

```
500.00
3
T1 C 100.00
T2 D 250.00
T3 D 500.00
```

6. Sample Output

```
T1 POSTED
T2 POSTED
T3 REJECTED
Final balance: 350.00
```